SECTION 1. IDENTIFICATION

Product name : AMBLISA®

Manufacturer or supplier's details

Company : FMC COLOMBIA S.A.S

Address : CALLE 108 # 45 30. TORRE 2,
OF. 1004-1005
BOGOTÁ D.C - COLOMBIA
+571 635150

Emergency telephone : 1 703 / 741-5970 (CHEMTREC - International)
+55 11 4349 1359 (CHEMTREC); +57 601 7942539 (CHEMTREC Bogota)
Colombia: 911

Medical Emergency Number : Desde Bogotá: 288 60 12; Línea Nacional: 01 8000 916012
Desde Ecuador: 1800 593005 (Quito, La Sierra, Centro y Norte).
Desde Perú: SAMU: 106;
CISPROQUIM®: 080-050-847;
FMC LATINOAMERICA S.A. SUCURSAL: 421-4811;
Desde Venezuela: 0800 1005012

Recommended use of the chemical and restrictions on use

Recommended use : Can be used as fungicide only.
Fungicide

Restrictions on use : Use as recommended by the label.

SECTION 2. HAZARDS IDENTIFICATION

GHS Classification

Acute toxicity (Oral) : Category 4

Acute toxicity (Inhalation) : Category 4

Acute toxicity (Dermal) : Category 4

Specific target organ toxicity - single exposure : Category 3 (Respiratory system)

Short-term (acute) aquatic hazard : Category 2

Long-term (chronic) aquatic hazard : Category 2

GHS label elements

Hazard pictograms :


Signal Word : WARNING

Hazard Statements : H302 + H312 + H332 Harmful if swallowed, in contact with skin or if inhaled.
H335 May cause respiratory irritation.
H411 Toxic to aquatic life with long lasting effects.

Precautionary Statements :

Prevention:

P261 Avoid breathing mist or vapors.
P264 Wash skin thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P271 Use only outdoors or in a well-ventilated area.
P273 Avoid release to the environment.
P280 Wear protective gloves/ protective clothing.

Response:

P301 + P312 + P330 IF SWALLOWED: Call a POISON CENTER/ doctor if you feel unwell. Rinse mouth.
P302 + P352 + P312 IF ON SKIN: Wash with plenty of water.Call a POISON CENTER/ doctor if you feel unwell.
P304 + P340 + P312 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER/ doctor if you feel unwell.
P362 + P364 Take off contaminated clothing and wash it before reuse.
P391 Collect spillage.

Storage:

P403 + P233 Store in a well-ventilated place. Keep container tightly closed.
P405 Store locked up.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

Hazard Statements required by Andean Technical Manual for the Registration and Control of Chemical Pesticides for Agricultural Use (Resolution no. 2075):
Harmful in contact with skin.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Chemical name	CAS-No.	Concentration (% w/w)

Flutriafol	76674-21-0	>= 25 -< 30
Fluindapyr	1383809-87-7	>= 10 -< 20
Poly(oxy-1,2-ethanediyl), α-tridecyl-ω-hydroxy-, phosphate, potassium salt	68186-36-7	>= 1 -< 2,5
Alkylated Naphthalene Sulfonate Sodium Salt	68425-94-5	>= 1 -< 2,5
1,2-benzisothiazol-3(2H)-one	2634-33-5	>= 0,0025 -< 0,025

SECTION 4. FIRST AID MEASURES

General advice : Move out of dangerous area.
Show this material safety data sheet to the doctor in attendance.
Do not leave the victim unattended.

If inhaled : If unconscious, place in recovery position and seek medical advice.
If symptoms persist, call a physician.

In case of skin contact : Take off contaminated clothing and shoes immediately.
Wash off with soap and water.
If symptoms persist, call a physician.
Wash contaminated clothing before re-use.

In case of eye contact : Flush eyes with water as a precaution.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.

If swallowed : Keep respiratory tract clear.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.

Most important symptoms and effects, both acute and delayed : Harmful if swallowed, in contact with skin or if inhaled.
May cause respiratory irritation.
Contains a triazole. Symptoms may include nausea, vomiting, diarrhea, visual changes, hallucinations, rash, itching, and alopecia.

Protection of first-aiders : Avoid inhalation, ingestion and contact with skin and eyes.

Notes to physician : Treat symptomatically.

SECTION 5. FIRE-FIGHTING MEASURES

Suitable extinguishing media : Dry chemical, CO_2, water spray or regular foam.

Unsuitable extinguishing media : Do not spread spilled material with high-pressure water streams.

Specific hazards during fire fighting : Do not allow run-off from fire fighting to enter drains or water courses.

Hazardous combustion products	:	Fire may produce irritating, corrosive and/or toxic gases. Hydrogen fluoride Nitrogen oxides (NOx) Carbon oxides Fluorinated compounds Hydrogen cyanide Sulfur oxides Fluorine compounds
Specific extinguishing methods	:	Remove undamaged containers from fire area if it is safe to do so. Use a water spray to cool fully closed containers. Standard procedure for chemical fires. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Special protective equipment for fire-fighters	:	Firefighters should wear protective clothing and self-contained breathing apparatus.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	:	Evacuate personnel to safe areas. Use personal protective equipment. If it can be safely done, stop the leak. Do not touch or walk through the spilled material.
Environmental precautions	:	Prevent product from entering drains. Prevent further leakage or spillage if safe to do so. If the product contaminates rivers and lakes or drains inform respective authorities.
Methods and materials for containment and cleaning up	:	Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Never return spills in original containers for re-use.

SECTION 7. HANDLING AND STORAGE

Advice on protection against fire and explosion	:	Normal measures for preventive fire protection.
Advice on safe handling	:	Avoid formation of aerosol. Do not breathe vapors/dust. Avoid exposure - obtain special instructions before use. For personal protection see section 8. Smoking, eating and drinking should be prohibited in the application area. Provide sufficient air exchange and/or exhaust in work rooms. Dispose of rinse water in accordance with local and national regulations.

Conditions for safe storage	:	Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage. Electrical installations / working materials must comply with the technological safety standards.
Further information on storage stability	:	No decomposition if stored and applied as directed.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Contains no substances with occupational exposure limit values.

Personal protective equipment

Respiratory protection	:	In the case of dust or aerosol formation use respirator with an approved filter.
Hand protection		
Material	:	Protective gloves
Remarks	:	The suitability for a specific workplace should be discussed with the producers of the protective gloves.
Eye protection	:	Eye wash bottle with pure water Tightly fitting safety goggles
Skin and body protection	:	Impervious clothing Choose body protection according to the amount and concentration of the dangerous substance at the work place.
Protective measures	:	Plan first aid action before beginning work with this product.
Hygiene measures	:	Wash hands before breaks and immediately after handling the product. Avoid contact with skin, eyes and clothing. Do not inhale aerosol.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	:	liquid
Form	:	liquid
Color	:	off-white
Odor	:	No data available

Odor Threshold	:	No data available
pH	:	6,69
Melting point/ range	:	No data available
Boiling point/boiling range	:	No data available
Flash point	:	> 100 °C
Self-ignition	:	No data available
Upper explosion limit / Upper flammability limit	:	No data available
Lower explosion limit / Lower flammability limit	:	No data available
Vapor pressure	:	No data available
Density	:	1,147 g/cm3
Solubility(ies)		
Water solubility	:	No data available
Partition coefficient: n-octanol/water	:	No data available
Autoignition temperature	:	No data available
Decomposition temperature	:	No data available
Viscosity		
Viscosity, dynamic	:	No data available
Viscosity, kinematic	:	No data available
Explosive properties	:	Not explosive
Oxidizing properties	:	Non-oxidizing

SECTION 10. STABILITY AND REACTIVITY

Reactivity	:	No decomposition if stored and applied as directed.
Chemical stability	:	No decomposition if stored and applied as directed.
Possibility of hazardous reactions	:	No decomposition if stored and applied as directed.

Conditions to avoid	:	Avoid extreme temperatures. Avoid formation of aerosol.
Incompatible materials	:	Avoid strong acids, bases, and oxidizers.
Hazardous decomposition products	:	No hazardous decomposition products are known.

SECTION 11. TOXICOLOGICAL INFORMATION

Acute toxicity

Harmful if swallowed, in contact with skin or if inhaled.

Product:

Acute oral toxicity	:	LD50 (Rat): 1.098 mg/kg Assessment: The component/mixture is moderately toxic after single ingestion.
Acute inhalation toxicity	:	LC50 (Rat): > 2,07 mg/l Exposure time: 4 h Test atmosphere: dust/mist Assessment: The component/mixture is moderately toxic after short term inhalation.
Acute dermal toxicity	:	LD50 (Rat): > 5.000 mg/kg Assessment: The substance or mixture has no acute dermal toxicity Assessment: The component/mixture is moderately toxic after single contact with skin. Remarks: Resolution no. 2075

Components:

Flutriafol:

Acute oral toxicity	:	LD50 (Rat, female): 300 - 2.000 mg/kg Method: OECD Test Guideline 423 Target Organs: Liver, Gastrointestinal tract Symptoms: Fatality GLP: yes Assessment: The component/mixture is moderately toxic after single ingestion. LD50 (Rat, female): 1.030 mg/kg Method: OECD Test Guideline 425 Target Organs: Liver, Gastrointestinal tract Symptoms: Breathing difficulties
Acute inhalation toxicity	:	LC50 (Rat): > 5,2 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403

Symptoms: Fatality, ataxia, Breathing difficulties
GLP: yes

Acute dermal toxicity : LD50 (Rat, male and female): > 2.000 mg/kg
Method: OECD Test Guideline 402
GLP: yes
Remarks: no mortality

LD50 (Rat, male and female): > 5.000 mg/kg
Method: OECD Test Guideline 402
Symptoms: Irritation
GLP: yes
Assessment: The substance or mixture has no acute dermal toxicity
Remarks: no mortality

Fluindapyr:

Acute oral toxicity : LD50 (Rat, female): > 2.000 mg/kg
Method: OECD Test Guideline 423
GLP: yes
Assessment: The component/mixture is minimally toxic after single ingestion.
Remarks: no mortality

Acute inhalation toxicity : LC50 (Rat, male and female): > 5,19 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Symptoms: ataxia, Breathing difficulties
GLP: yes
Remarks: no mortality

Acute dermal toxicity : LD50 (Rat, male and female): > 2.000 mg/kg
Method: OECD Test Guideline 402
Symptoms: Irritation
GLP: yes
Assessment: The component/mixture is minimally toxic after single contact with skin.
Remarks: no mortality

Poly(oxy-1,2-ethanediyl), α-tridecyl-ω-hydroxy-, phosphate, potassium salt:

Acute oral toxicity : Assessment: Toxic effects cannot be excluded

Alkylated Naphthalene Sulfonate Sodium Salt:

Acute oral toxicity : LD50 (Rat): > 5.000 mg/kg

1,2-benzisothiazol-3(2H)-one:

Acute dermal toxicity : LD50 (Rat, male and female): > 2.000 mg/kg
Method: OECD Test Guideline 402
Assessment: The substance or mixture has no acute dermal toxicity

Skin corrosion/irritation

Based on available data, the classification criteria are not met.

Product:

Assessment : No skin irritation
Result : No skin irritation

Components:

Flutriafol:

Species : Rabbit
Assessment : Not classified as irritant
Method : OECD Test Guideline 404
Result : No skin irritation
GLP : yes

Species : Rabbit
Result : No skin irritation

Fluindapyr:

Species : Rabbit
Assessment : Not classified as irritant
Method : OECD Test Guideline 404
GLP : yes

Assessment : Not classified as irritant
Method : EPISKIN Human Skin Model Test

Poly(oxy-1,2-ethanediyl), α-tridecyl-ω-hydroxy-, phosphate, potassium salt:

Result : Skin irritation

Alkylated Naphthalene Sulfonate Sodium Salt:

Remarks : No data available

1,2-benzisothiazol-3(2H)-one:

Species : Rabbit
Exposure time : 72 h
Method : OECD Test Guideline 404
Result : No skin irritation

Serious eye damage/eye irritation

Based on available data, the classification criteria are not met.

Product:

Result : No eye irritation
Assessment : No eye irritation

Components:

Flutriafol:

Species : Rabbit
Result : slight irritation
Assessment : Not classified as irritant
Method : OECD Test Guideline 405
GLP : yes

Species : Rabbit
Result : Slight or no eye irritation
Assessment : Not classified as irritant
Method : OECD Test Guideline 405
GLP : yes

Species : Rabbit
Result : slight irritation

Fluindapyr:

Species : Rabbit
Result : No eye irritation
Method : OECD Test Guideline 405
GLP : yes

Result : not corrosive
Method : Bovine cornea (BCOP)
GLP : yes

Poly(oxy-1,2-ethanediyl), α-tridecyl-ω-hydroxy-, phosphate, potassium salt:

Result : Irreversible effects on the eye

Alkylated Naphthalene Sulfonate Sodium Salt:

Result : Eye irritation

1,2-benzisothiazol-3(2H)-one:

Species : Bovine cornea
Result : No eye irritation
Method : OECD Test Guideline 437

Species : Rabbit
Result : Irreversible effects on the eye
Method : EPA OPP 81-4

Respiratory or skin sensitization

Skin sensitization

Based on available data, the classification criteria are not met.

Respiratory sensitization

Based on available data, the classification criteria are not met.

Product:

Assessment	:	Not a skin sensitizer.
Result	:	Not a skin sensitizer.

Components:

Flutriafol:

Test Type	:	Buehler Test
Routes of exposure	:	Skin contact
Species	:	Guinea pig
Assessment	:	Did not cause sensitization on laboratory animals.
Method	:	OECD Test Guideline 406
GLP	:	yes
Species	:	Guinea pig
Result	:	Not a skin sensitizer.

Fluindapyr:

Test Type	:	Local lymph node assay (LLNA)
Routes of exposure	:	Skin contact
Method	:	OECD Test Guideline 429
Result	:	May cause sensitization by skin contact.
GLP	:	yes

1,2-benzisothiazol-3(2H)-one:

Test Type	:	Maximization Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	May cause sensitization by skin contact.
Species	:	Guinea pig
Method	:	FIFRA 81.06
Result	:	May cause sensitization by skin contact.

Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Components:

Flutriafol:

Genotoxicity in vivo	:	Test Type: dominant lethal test Method: OECD Test Guideline 478 Result: negative

Fluindapyr:

Genotoxicity in vitro	:	Test Type: Chromosome aberration test in vitro Test system: lymphocytes Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 473 Result: negative Test Type: gene mutation test

		Test system: mouse lymphoma cells Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 490 Result: negative
		Test Type: Ames test Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 471 Result: negative
		Test Type: In vitro mammalian cell gene mutation test Test system: mouse lymphoma cells Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 476 Result: negative GLP: yes
Genotoxicity in vivo	:	Test Type: Mammalian bone marrow sister chromatid exchange Species: Mouse Result: negative
		Test Type: Micronucleus test Species: Mouse Method: OECD Test Guideline 474 Result: negative

1,2-benzisothiazol-3(2H)-one:

Genotoxicity in vitro	:	Test Type: gene mutation test Test system: mouse lymphoma cells Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 476 Result: negative
		Test Type: Ames test Method: OECD Test Guideline 471 Result: negative
		Test Type: Chromosome aberration test in vitro Method: OECD Test Guideline 473 Result: positive
Genotoxicity in vivo	:	Test Type: unscheduled DNA synthesis assay Species: Rat (male) Cell type: Liver cells Application Route: Ingestion Exposure time: 4 h Method: OECD Test Guideline 486 Result: negative
		Test Type: Micronucleus test Species: Mouse Application Route: Oral Method: OECD Test Guideline 474

		Result: negative
Germ cell mutagenicity - Assessment	:	Weight of evidence does not support classification as a germ cell mutagen.

Carcinogenicity

Based on available data, the classification criteria are not met.

Components:

Flutriafol:

Species	:	Mouse
Exposure time	:	2 Years
NOAEL	:	1,2 mg/kg bw/day
Result	:	negative
Species	:	Rat
Exposure time	:	2 Years
NOAEL	:	1 mg/kg bw/day
Result	:	negative
Carcinogenicity - Assessment	:	Animal testing did not show any carcinogenic effects.

Fluindapyr:

Species	:	Mouse
Application Route	:	Oral
Exposure time	:	18 month(s)
Method	:	OECD Test Guideline 451
Result	:	Not a carcinogenic hazard
Species	:	Rat
Application Route	:	Oral
Exposure time	:	2 Years
Method	:	OECD Test Guideline 453
Result	:	Not a carcinogenic hazard
GLP	:	yes

Reproductive toxicity

Based on available data, the classification criteria are not met.

Components:

Flutriafol:

Effects on fertility	:	Test Type: reproductive and developmental toxicity study Method: OECD Test Guideline 416 Result: negative
Effects on fetal development	:	Test Type: Embryo-fetal development Method: OECD Test Guideline 414 Result: negative

Fluindapyr:

Effects on fertility	:	Test Type: Two-generation study General Toxicity Parent: NOEL: ca. 100 ppm Fertility: NOAEL: ca. 400 ppm Early Embryonic Development: NOAEL: ca. 400 ppm Method: OECD Test Guideline 416 GLP: yes

1,2-benzisothiazol-3(2H)-one:

Effects on fertility	:	Species: Rat, male Application Route: Ingestion General Toxicity Parent: NOAEL: 18,5 mg/kg body weight General Toxicity F1: NOAEL: 48 mg/kg body weight Fertility: NOAEL: 112 mg/kg bw/day Symptoms: No effects on reproduction parameters. Method: OPPTS 870.3800 Result: negative
Reproductive toxicity - Assessment	:	Weight of evidence does not support classification for reproductive toxicity

STOT-single exposure

May cause respiratory irritation.

Components:

Flutriafol:

Assessment	:	May cause respiratory irritation.

STOT-repeated exposure

Based on available data, the classification criteria are not met.

Components:

1,2-benzisothiazol-3(2H)-one:

Assessment	:	The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Repeated dose toxicity

Components:

Flutriafol:

Species	:	Rat
NOAEL	:	13.3 mg/kg bw/day
Application Route	:	Oral - feed
Exposure time	:	90 d
Symptoms	:	anemia, Liver effects
Species	:	Dog
NOAEL	:	5 mg/kg bw/day
Application Route	:	Oral
Exposure time	:	90 d
Symptoms	:	blood effects, Liver effects

Fluindapyr:

Species	:	Rat
NOAEL	:	1.000 mg/kg
Application Route	:	Dermal
Exposure time	:	21 d
Number of exposures	:	5 d/w for 6 hr
Dose	:	0,100,300,1000 mg/kg bw/d
Method	:	OECD Test Guideline 410
GLP	:	yes
Symptoms	:	Skin irritation

1,2-benzisothiazol-3(2H)-one:

Species	:	Rat, male and female
NOAEL	:	15 mg/kg
Application Route	:	Ingestion
Exposure time	:	28 d
Method	:	OECD Test Guideline 407
Symptoms	:	Irritation

Species	:	Rat, male and female
NOAEL	:	69 mg/kg
Application Route	:	Ingestion
Exposure time	:	90 d
Symptoms	:	Irritation, Reduced body weight

Aspiration toxicity

Based on available data, the classification criteria are not met.

Components:

Flutriafol:

The substance does not have properties associated with aspiration hazard potential.

Neurological effects

Components:

Flutriafol:

No neurotoxicity observed in animal studies.

Further information

Product:

Remarks	:	No data available

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity

Product:

Ecotoxicology Assessment

Acute aquatic toxicity : Toxic to aquatic life.

Chronic aquatic toxicity : Toxic to aquatic life with long lasting effects.

Components:

Flutriafol:

Toxicity to fish : LC50 (Lepomis macrochirus (Bluegill sunfish)): 33 mg/l
Exposure time: 96 h

LC50 (Danio rerio (zebra fish)): 22,97 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 67 mg/l
End point: Immobilization
Exposure time: 48 h
Test Type: static test
Method: OECD Test Guideline 202
GLP: yes

EC50 (Daphnia magna (Water flea)): 42,21 mg/l
End point: Immobilization
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : IC50 (Selenastrum capricornutum (green algae)): 12 mg/l
Exposure time: 96 h

IC50 (Scenedesmus subspicatus): 1,9 mg/l
Exposure time: 72 h

EbC50 (Lemna gibba (duckweed)): 0,65 mg/l
Exposure time: 7 d

EyC50 (Pseudokirchneriella subcapitata (algae)): 3,69 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201
GLP: yes

Toxicity to fish (Chronic toxicity) : NOEC (Lepomis macrochirus (Bluegill sunfish)): 4,8 mg/l
Exposure time: 28 d

NOEC (Danio rerio (zebra fish)): 20 mg/l
Exposure time: 14 d
Method: OECD Test Guideline 204

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC (Daphnia magna (Water flea)): 0,31 mg/l Exposure time: 21 d
		NOEC (Daphnia magna (Water flea)): 0,45 mg/l Exposure time: 21 d Method: OECD Test Guideline 211
Toxicity to soil dwelling organisms	:	NOEC (Eisenia fetida (earthworms)): 0.01 mg/cm2 Exposure time: 180 d
		LC50 (Eisenia fetida (earthworms)): > 1.000 mg/kg Exposure time: 14 d Method: OECD Test Guideline 207
Toxicity to terrestrial organisms	:	LD50 (Apis mellifera (bees)): > 144 µg/bee End point: Acute oral toxicity Method: OECD Test Guideline 213 GLP: yes
		LD50 (Apis mellifera (bees)): > 150 µg/bee End point: Acute contact toxicity Method: OECD Test Guideline 214 GLP: yes
		LD50 (Apis mellifera (bees)): > 100 µg/bee End point: Acute contact toxicity Method: OECD Test Guideline 214
		LD50 (Apis mellifera (bees)): 872,53 µg/bee Exposure time: 48 h End point: Acute oral toxicity Method: OECD Test Guideline 213
		LD50 (Anas platyrhynchos (Mallard duck)): > 5.000 mg/kg
		LD50 (Coturnix japonica (Japanese quail)): ca. 385 mg/kg Method: US EPA Test Guideline OPPTS 850.2100
		LD50 (Coturnix japonica (Japanese quail)): 4260 ppm Method: OPPTS 850.2200

Fluindapyr:

Toxicity to fish	:	LC50 (Oncorhynchus mykiss (rainbow trout)): 0,121 mg/l Exposure time: 96 h Test Type: static test Method: OECD Test Guideline 203 GLP: yes
		LC50 (Oryzias latipes (Japanese medaka)): > 1,8 mg/l Exposure time: 96 h Test Type: static test Method: OECD Test Guideline 203 GLP: yes

		LC50 (Danio rerio (zebra fish)): 0,424 mg/l Exposure time: 96 h Test Type: static test Method: OECD Test Guideline 203 GLP: yes
		LC50 (Cyprinodon variegatus (sheepshead minnow)): 0,43 mg/l Exposure time: 96 h Test Type: static test Method: OPPTS 850.1075 GLP: yes
		LC50 (Cyprinus carpio (Carp)): 0,11 mg/l Exposure time: 96 h Test Type: Static renewal test Method: OECD Test Guideline 203 GLP: yes
		LC50 (Lepomis macrochirus (Bluegill sunfish)): 0,286 mg/l Exposure time: 96 h Test Type: static test Method: OECD Test Guideline 203 GLP: yes
		LC50 (Pimephales promelas (fathead minnow)): 0,19 mg/l Exposure time: 96 h Method: OECD Test Guideline 203 GLP: yes
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 0,141 mg/l Exposure time: 48 h Test Type: static test Method: OECD Test Guideline 202
		LC50 (Americamysis bahia (mysid shrimp)): 0,33 mg/l Exposure time: 96 h Test Type: static test Method: OCSPP 850.1035 GLP: yes
Toxicity to algae/aquatic plants	:	ErC50 (Pseudokirchneriella subcapitata (green algae)): > 4,83 mg/l Exposure time: 72 h Method: OECD Test Guideline 201 GLP: yes
		NOEC (Lemna gibba (duckweed)): 2 mg/l Exposure time: 7 d Method: OECD Test Guideline 221 GLP: yes
		EC50 (Skeletonema costatum (Diatom)): > 2 mg/l Exposure time: 72 h

		Method: OECD Test Guideline 201 GLP: yes
M-Factor (Acute aquatic toxicity)	:	1
Toxicity to fish (Chronic toxicity)	:	NOEC (Pimephales promelas (fathead minnow)): 0,031 mg/l Exposure time: 32 d Test Type: Early-life Stage Method: OECD Test Guideline 210 GLP: yes
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC (Americamysis bahia (mysid shrimp)): 0,062 mg/l Exposure time: 28 d Test Type: flow-through test Method: OPPTS 850.1350 GLP: yes NOEC (Daphnia magna (Water flea)): 0,12 mg/l End point: reproduction Exposure time: 21 d Method: OECD Test Guideline 211 GLP: yes NOEC (Hyalella azteca (Amphipod)): 68 mg/l Exposure time: 42 d Test Type: Static renewal test GLP: yes
M-Factor (Chronic aquatic toxicity)	:	1
Toxicity to soil dwelling organisms	:	LC50 (Eisenia fetida (earthworms)): > 1.000 mg/kg Method: OECD Test Guideline 216 Remarks: No significant adverse effect on Nitrogen mineralization. Method: OECD Test Guideline 217 Remarks: No significant adverse effect on Carbon mineralization.
Toxicity to terrestrial organisms	:	LD50 (Colinus virginianus (Bobwhite quail)): 1.612 mg/kg Method: OECD Test Guideline 205 GLP: yes LD50 (Colinus virginianus (Bobwhite quail)): 2.250 mg/kg Method: OPPTS 850.2100 GLP: yes LD50 (Apis mellifera (bees)): > 300 µg/bee Exposure time: 48 h Method: OECD Test Guideline 214 GLP: yes

Remarks: Contact

LD50 (Apis mellifera (bees)): > 32,8 µg/bee
Exposure time: 48 h
Method: OECD Test Guideline 213
GLP: yes
Remarks: Oral

NOEC (Anas platyrhynchos (Mallard duck)): 679 mg/kg
End point: Reproduction Test
Method: OECD Test Guideline 206
GLP: yes

NOEC (Colinus virginianus (Bobwhite quail)): 174 mg/kg
End point: Reproduction Test
Method: OECD Test Guideline 206
GLP: yes

Poly(oxy-1,2-ethanediyl), α-tridecyl-ω-hydroxy-, phosphate, potassium salt:

Ecotoxicology Assessment

Acute aquatic toxicity : Harmful to aquatic life.

Chronic aquatic toxicity : Harmful to aquatic life with long lasting effects.

Alkylated Naphthalene Sulfonate Sodium Salt:

Toxicity to fish : LC50 (Zebra fish): > 10 - 100 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203
Remarks: Based on data from similar materials

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 100 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202
Remarks: Based on data from similar materials

Toxicity to algae/aquatic plants : EC50 (Pseudokirchneriella subcapitata (green algae)): > 100 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201
Remarks: Based on data from similar materials

EC10 (Pseudokirchneriella subcapitata (green algae)): > 100 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201
Remarks: Based on data from similar materials

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : EC10 (Daphnia magna (Water flea)): > 10 - 100 mg/l
Exposure time: 21 d
Method: OECD Test Guideline 211
Remarks: Based on data from similar materials

1,2-benzisothiazol-3(2H)-one:

Toxicity to fish	:	LC50 (Cyprinodon variegatus (sheepshead minnow)): 16,7 mg/l Exposure time: 96 h Test Type: static test LC50 (Oncorhynchus mykiss (rainbow trout)): 2,15 mg/l Exposure time: 96 h Method: OECD Test Guideline 203
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 2,9 mg/l Exposure time: 48 h Test Type: static test Method: OECD Test Guideline 202
Toxicity to algae/aquatic plants	:	EC50 (Pseudokirchneriella subcapitata (green algae)): 0,070 mg/l Exposure time: 72 h Method: OECD Test Guideline 201 NOEC (Pseudokirchneriella subcapitata (green algae)): 0,04 mg/l Exposure time: 72 h Method: OECD Test Guideline 201
M-Factor (Acute aquatic toxicity)	:	10
Toxicity to microorganisms	:	EC50 (activated sludge): 24 mg/l Exposure time: 3 h Test Type: Respiration inhibition Method: OECD Test Guideline 209 EC50 (activated sludge): 12,8 mg/l Exposure time: 3 h Test Type: Respiration inhibition Method: OECD Test Guideline 209

Persistence and degradability

Components:

Flutriafol:

Biodegradability	:	Result: Not readily biodegradable.
Stability in water	:	Remarks: Does not readily hydrolyze

Fluindapyr:

Biodegradability	:	Result: Not readily biodegradable.

Poly(oxy-1,2-ethanediyl), α-tridecyl-ω-hydroxy-, phosphate, potassium salt:

Biodegradability	:	Result: Readily biodegradable. Biodegradation: 80 %

Exposure time: 28 d
Method: OECD Test Guideline 301D
Remarks: Based on data from similar materials

Alkylated Naphthalene Sulfonate Sodium Salt:

Biodegradability : Result: Not readily biodegradable.
Remarks: Based on data from similar materials

1,2-benzisothiazol-3(2H)-one:

Biodegradability : Result: rapidly biodegradable
Method: OECD Test Guideline 301C

Bioaccumulative potential

Components:

Flutriafol:

Bioaccumulation : Species: Fish
Bioconcentration factor (BCF): 7
Remarks: Bioaccumulation is unlikely.

Partition coefficient: n-octanol/water : log Pow: 2,29

Fluindapyr:

Bioaccumulation : Species: Lepomis macrochirus (Bluegill sunfish)
Bioconcentration factor (BCF): < 500
Method: OECD Test Guideline 305
GLP: yes
Remarks: Bioaccumulation is unlikely.

Partition coefficient: n-octanol/water : log Pow: > 3

1,2-benzisothiazol-3(2H)-one:

Bioaccumulation : Species: Lepomis macrochirus (Bluegill sunfish)
Bioconcentration factor (BCF): 6,62
Exposure time: 56 d
Method: OECD Test Guideline 305
Remarks: Substance is not persistent, bioaccumulative, and toxic (PBT).

Partition coefficient: n-octanol/water : log Pow: 0,7 (20 °C)
pH: 7

log Pow: 0,99 (20 °C)
pH: 5

Mobility in soil

Components:

Flutriafol:

Distribution among environmental compartments : Remarks: Moderately mobile in soils

Stability in soil : Remarks: Very persistent in soil.

Fluindapyr:

Distribution among environmental compartments : Remarks: Low mobility in soil.

1,2-benzisothiazol-3(2H)-one:

Distribution among environmental compartments : Koc: 9,33 ml/g, log Koc: 0,97
Method: OECD Test Guideline 121
Remarks: Highly mobile in soils

Other adverse effects

Product:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Toxic to aquatic life with long lasting effects.

Components:

Flutriafol:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Toxic to aquatic life with long lasting effects.

Global warming potential

Assessment Report of the Intergovernmental Panel on Climate Change (IPCC) of the United Nations Framework Convention on Climate Change (UNFCCC)

Components:

octamethylcyclotetrasiloxane [D4]:

20-year global warming potential: 2,66
100-year global warming potential: 0,739
500-year global warming potential: 0,211
Atmospheric lifetime: 0,027 yr
Radiative efficiency: 0,12 Wm2ppb
Further information: Miscellaneous compounds

decamethylcyclopentasiloxane:

20-year global warming potential: 1,04
100-year global warming potential: 0,289
500-year global warming potential: 0,082

Atmospheric lifetime: 0,016 yr
Radiative efficiency: 0,098 Wm2ppb
Further information: Miscellaneous compounds

dodecamethylcyclohexasiloxane:

20-year global warming potential: 0,51
100-year global warming potential: 0,142
500-year global warming potential: 0,04
Atmospheric lifetime: 0,011 yr
Radiative efficiency: 0,086 Wm2ppb
Further information: Miscellaneous compounds

SECTION 13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues	:	The product should not be allowed to enter drains, water courses or the soil. Do not contaminate ponds, waterways or ditches with chemical or used container. Send to a licensed waste management company.
Contaminated packaging	:	Empty remaining contents. Dispose of as unused product. Do not re-use empty containers. Empty containers should be taken to an approved waste handling site for recycling or disposal.

SECTION 14. TRANSPORT INFORMATION

International Regulations

UNRTDG

UN number	:	UN 3082
Proper shipping name	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Fluindapyr, Flutriafol)
Class	:	9
Packing group	:	III
Labels	:	9
Environmentally hazardous	:	yes

IATA-DGR

UN/ID No.	:	UN 3082
Proper shipping name	:	Environmentally hazardous substance, liquid, n.o.s. (Fluindapyr, Flutriafol)
Class	:	9
Packing group	:	III
Labels	:	Miscellaneous
Packing instruction (cargo aircraft)	:	964

Packing instruction (passenger aircraft) : 964
Environmentally hazardous : yes

IMDG-Code
UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Fluindapyr, Flutriafol)

Class : 9
Packing group : III
Labels : 9
EmS Code : F-A, S-F
Marine pollutant : yes

Transport in bulk according to IMO instruments

Not applicable for product as supplied.

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

SECTION 15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Substances and chemicals controlled by the Ministry of Justice : Neither banned nor restricted

List of substances included for special control and subject to supervision by the Ministry of Health and Social Protection : Not applicable

Resolution 2715/2014, which establishes the substances subject to registration of retail sales, based on defined classification criteria. : Not applicable

The ingredients of this product are reported in the following inventories:

TCSI : Not in compliance with the inventory

TSCA : Product contains substance(s) not listed on TSCA inventory.

AIIC : Not in compliance with the inventory

DSL : This product contains the following components that are not on the Canadian DSL nor NDSL.

Flutriafol
Fluindapyr
Smectite-group minerals

ENCS : Not in compliance with the inventory

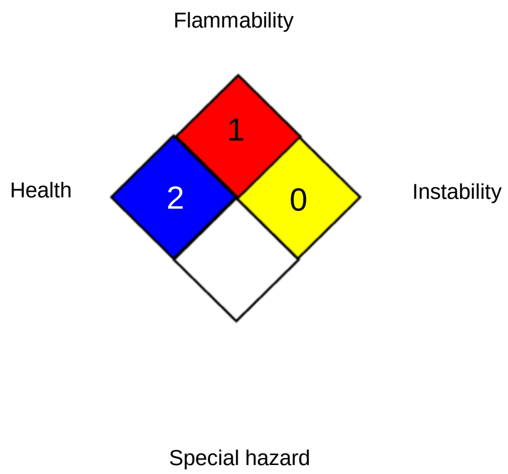
ISHL	:	Not in compliance with the inventory
KECI	:	Not in compliance with the inventory
PICCS	:	Not in compliance with the inventory
IECSC	:	Not in compliance with the inventory
NZIoC	:	Not in compliance with the inventory
TECI	:	Not in compliance with the inventory

SECTION 16. OTHER INFORMATION

Revision Date	:	02.02.2026
Date format	:	dd.mm.yyyy

Further information

NFPA:



HMIS® IV:

HEALTH	/	1
FLAMMABILITY		1
PHYSICAL HAZARD		0

HMIS® ratings are based on a 0-4 rating scale, with 0 representing minimal hazards or risks, and 4 representing significant hazards or risks. The "*" represents a chronic hazard, while the "/" represents the absence of a chronic hazard.

Full text of other abbreviations

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and

Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

Disclaimer

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